

Curriculum Vitæ — Peter Haine

University of Southern California
Kaprielian Hall, 248C
Los Angeles, CA 90089

Website peterjhaine.github.io

E-mail phaine@usc.edu

Citizenship *United States*

Employment

Fall 2025– **Assistant Professor**, University of Southern California

2023–2025 **NSF Postdoctoral Fellow**, UC Berkeley
Sponsoring scientist: David Nadler

2022–2023 **Member**, Institute for Advanced Study
Sponsoring scientist: Jacob Lurie

2021–2022 **NSF Postdoctoral Fellow and UC President’s Postdoctoral Fellow**, UC Berkeley
Sponsoring scientist: David Nadler

Education

2021 **Massachusetts Institute of Technology**, Cambridge, MA, *Ph.D. Mathematics*
Advisor: Clark Barwick

2016 **Massachusetts Institute of Technology**, Cambridge, MA, *S.B. Mathematics*

Research Interests

Homotopy theory, algebraic geometry, & related subjects

Short-term Visits

June 2025 **IMPAN Gdańsk**, Sopot, Poland

March 2025 **Copenhagen Centre for Geometry and Topology**, Copenhagen, Denmark

July 2024 **Institut de Recherche Mathématiques Avancées**, Strasbourg, France

June 2024 **Copenhagen Centre for Geometry and Topology**, Copenhagen, Denmark

June 2023 **Max-Planck-Institut für Mathematik**, Bonn, Germany

June–July 2022 **Universität Regensburg**, Regensburg, Germany

June 2022 **Higher Categories and Categorification, Part 2**, MSRI/Unidad Cuernavaca del Instituto de Matemáticas UNAM, Cuernavaca, México

Spring 2020 **Higher Categories and Categorification**, MSRI, Berkeley, CA

Selected Awards & Distinctions

2021–2025	National Science Foundation Mathematical Sciences Postdoctoral Research Fellowship NSF Award DMS-2102957
2021–2022	President’s Postdoctoral Fellowship, University of California
2021	Mathematics Community Building Award, MIT
2020–2021	George Lusztig PRIMES Mentorship, MIT
Fall 2019	AMS Graduate Student Travel Grant
August 2019	Leibniz Graduate Student Grant, Mathematisches Forschungsinstitut Oberwolfach Workshop on <i>Homotopy Theory</i>
2016–2021	National Science Foundation Graduate Research Fellowship
2016–2019	Dean of Science Fellow, MIT

Publications & Preprints

17. P. J. Haine, T. Holzschuh, M. Lara, C. Mair, L. Martini, and S. Wolf, *The condensed homotopy type of a scheme*, Oct. 2025, With an appendix by B. Zavyalov. [arXiv:2510.07443](#).
16. P. J. Haine, M. Porta, and J.-B. Teyssier, *Standard t-structures*, Apr. 2025, [arXiv:2504.07473](#).
15. P. J. Haine, M. Ramzi, and J. Steinebrunner, *Fully faithful functors and pushouts of ∞ -categories*, Mar. 2025, [arXiv:2503.03916](#).
14. M. Carlson, P. J. Haine, and S. Wolf, *A generalization of Voevodsky’s reconstruction theorem of schemes from their étale topoi*, *Selecta Math. (N.S.)*, vol. 31, no. 4, Paper No. 65, 37, 2025. DOI: 10.1007/s00029-025-01054-5, [arXiv:2407.19920](#).
13. P. J. Haine, M. Porta, and J.-B. Teyssier, *Exodromy beyond conicality*, Jan. 2024, [arXiv:2401.12825](#).
12. P. J. Haine and P. Pstrągowski, *Spectral weight filtrations*, Sep. 2023, [arXiv:2309.15072](#).
11. P. J. Haine, T. Holzschuh, and S. Wolf, *Nonabelian basechange theorems and étale homotopy theory*, *J. Topol.*, vol. 17, no. 4, Paper No. e70009, 45, 2024. DOI: 10.1112/topo.70009, [arXiv:2304.00938](#).
10. P. J. Haine, *Profinite completions of products*, May 2024, To appear in *Algebraic and Geometric Topology*. [arXiv:2406.00136](#).
9. P. J. Haine, T. Holzschuh, and S. Wolf, *The fundamental fiber sequence in étale homotopy theory*, *Int. Math. Res. Not.*, no. 1, pp. 175–196, 2024. DOI: 10.1093/imrn/rnad018, [arXiv:2209.03476](#).
8. A. Amabel, A. Debray, and P. J. Haine, Eds., *Differential cohomology: Categories, characteristic classes, and connections*, Jan. 2023, [arXiv:2109.12250](#). To be published in *Cambridge Studies in Advanced Mathematics*.
7. P. J. Haine, *From nonabelian basechange to basechange with coefficients*, *J. Pure Appl. Algebra*, vol. 229, no. 7, Paper No. 107993, 30, 2025. DOI: 10.1016/j.jpaa.2025.107993, [arXiv:2108.03545](#).
6. P. J. Haine, M. Porta, and J.-B. Teyssier, *The homotopy-invariance of constructible sheaves*, *Homology Homotopy Appl.*, vol. 25, no. 2, pp. 97–128, 2023. DOI: 10.4310/hha.2023.v25.n2.a6, [arXiv:2010.06473](#).
5. P. J. Haine, *On the homotopy theory of stratified spaces*, *Ann. Sci. Éc. Norm. Supér. (4)*, vol. 57, no. 2, pp. 351–384, 2024, [arXiv:1811.01119](#).
4. C. Barwick, S. Glasman, and P. J. Haine, *Exodromy*, Aug. 2020, [arXiv:1807.03281](#). Includes the contents of the four papers [arXiv:1812.11637](#), [arXiv:1901.09414](#), [arXiv:1904.01877](#), and [arXiv:1904.09966](#).

3. S. Devalapurkar and P. J. Haine, *On the James and Hilton–Milnor Splittings, and the metastable EHP sequence*, Doc. Math., vol. 26, pp. 1423–1464, 2021, arXiv:1912.04130.
2. V. Guillemin and P. Haine, *Differential forms*. World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ, 2019, pp. xvi+256, ISBN: 978-981-3272-77-4. DOI: 10.1142/11058.
1. S. Fomin, R. Shankar, P. Haine, and V. Chugunov, *Stability analysis of non-Newtonian rimming flow*, Appl. Math. Model., vol. 40, no. 4, pp. 2999–3010, 2016. DOI: 10.1016/j.apm.2015.09.088.

Lecture Series

- | | |
|--------------------|--|
| June 16–20, 2025 | Étale cohomology and étale homotopy , Goethe-Universität Frankfurt
<i>Étale homotopy theory and exodromy</i> |
| June 24–28, 2024 | Masterclass on Exit Paths and Stratified Homotopy Types , Københavns Universitet
<i>Exodromy beyond conicality</i> |
| August 16–20, 2023 | Symplectic Paradise Summer School , Paradise Valley, Montana
<i>From stratifications to singular Lagrangians</i> |

Selected Talks

- | | |
|--------------------|--|
| October 9, 2025 | Young Homotopy Theorist Meeting , ICMS, Edinburgh
<i>The condensed homotopy type of a scheme</i> |
| June 10, 2025 | Modern Developments in Geometry and Higher Structures , Brin Center, University of Maryland
<i>The proétale homotopy type</i> |
| January 12, 2025 | JMM Satellite Session on Homotopy Theory , University of Washington
<i>Applications of exodromy in topology</i> |
| November 12, 2024 | Algebraic Geometry Seminar , Johns Hopkins
<i>Reconstructing schemes from their étale topoi</i> |
| November 4, 2024 | Colloquium , University of Southern California
<i>Exit-path categories in topology and algebraic geometry</i> |
| October 28, 2024 | RTG Seminar on Arithmetic Geometry , UC Berkeley
<i>Weight filtrations on the Betti cohomology of complex varieties</i> |
| September 2024 | Anabelian and Motivic Homotopy Theory , Universität Heidelberg
<i>Reconstructing schemes from their étale topoi</i> |
| September 17, 2024 | Seminar on Commutative Algebra and Algebraic Geometry , UC Berkeley
<i>Reconstructing schemes from their étale topoi</i> |
| September 2024 | Toposes in Mondovì , Grothendieck Institute
<i>Reconstructing schemes from their étale topoi</i> |
| August 27, 2024 | Enumerative Geometry Beyond Spaces , Banff International Research Station
<i>Interactions between homotopy theory and algebraic geometry</i> |

May 9, 2024	Szygies and Mirror Symmetry Virtual Seminar <i>An introduction to “Exodromy beyond conicality”</i>
April 17, 2024	Algebraic Geometry Seminar, UC Davis <i>Exodromy in topology and applications</i>
March 17, 2024	Lloyd Roeling Conference, University of Louisiana at Lafayette <i>Exodromy in topology and applications</i>
December 12, 2023	Category Theory Seminar, Johns Hopkins <i>How to invent pyknotic/condensed mathematics</i>
December 11, 2023	Topology Seminar, Johns Hopkins <i>Exodromy beyond conicality</i>
November 6, 2023	Algebra Seminar, University of Southern California <i>On Voevodsky’s reconstruction theorem</i>
October 20, 2023	Algebra & Number Theory Seminar, University of California Santa Cruz <i>Weight filtrations from the perspective of homotopy theory</i>
June 20, 2023	Arbeitstagung 2023 on Condensed Mathematics, Max-Planck-Institut für Mathematik, Bonn <i>Condensed math & exodromy</i>
May 5, 2023	Algebraic Topology Seminar, Columbia <i>Spectral weight filtrations</i>
April 24, 2023	Topology Seminar, MIT <i>Spectral weight filtrations</i>
March 20, 2023	Mathematics Seminar, Montana State University <i>Reconstruction and étale homotopy theory</i>
February 16, 2023	Geometry-Topology Seminar, University of Pennsylvania <i>Reconstruction and étale homotopy theory</i>
January 13, 2023	Spring Opportunities Workshop, Institute for Advanced Study <i>Reconstruction in algebraic geometry</i>
October 18, 2022	Topology Seminar, University of California San Diego <i>New perspectives on the étale homotopy type</i>
July 28, 2022	Oberseminar: Higher Topoi and Exodromy, Universität Regensburg <i>Microlocal sheaves</i>
July 7, 2022	Derived Categories, Arithmetic, and Reconstruction in Algebraic Geometry, Banff International Research Station <i>Galois-theoretic reconstruction of schemes and exodromy</i>
May 13, 2022	Algebraic Topology Seminar, UCLA <i>New perspectives on the reconstruction of schemes and étale homotopy types</i>

May 6, 2022	Homotopy Theory Seminar, University of Pennsylvania <i>Revisiting classical splitting results</i>
March 28, 2022	MUSA Math Monday, UC Berkeley <i>The cobordism hypothesis in low dimensions</i>
February 2, 2022	Oberseminar: Higher Topoi and Exodromy, Universität Regensburg <i>Exit-path categories in algebra and geometry</i>
January 24, 2022	Algebra & Geometry Seminar, Caltech <i>Stratified étale homotopy theory</i>
January 11, 2022	Research Seminar on Algebraic Topology, Universität Hamburg <i>Exodromy for ℓ-adic sheaves (or how I got into condensed math)</i>
November 11, 2021	Algebraic & Geometric Topology Seminar, University of Nevada, Reno <i>Exit-path categories in geometry and topology</i>
October 30, 2021	Talbot 202One: Ambidexterity in Chromatic Homotopy Theory <i>Lecture 16: Character isomorphisms via tempered cohomology</i>
May 3, 2021	Topology Seminar, MIT <i>On the homotopy theory of stratified spaces</i>
February 11, 2021	Thursday Seminar: Redshift in algebraic K-theory, Harvard <i>Introduction to algebraic K-theory</i>
December 14, 2020	Research Seminar on Algebraic Topology, Universität Hamburg <i>Revisiting classical splitting results</i>
December 7, 2020	Algebraic Topology Seminar, University of Michigan <i>Stratified étale homotopy theory</i>
November 3, 2020	UChicago-Northwestern Topology Seminar <i>Stratified étale homotopy theory</i>
June 2, 2020	Oberseminar: Integral Homotopy Theory, Universität Regensburg <i>Borel Global Algebras</i>
May 20, 2020	Motives and What Not, Motivic Zoom Conference Series <i>Stratified étale homotopy theory</i>
May 5, 2020	SFB Lecture, Universität Regensburg <i>New perspectives on étale homotopy theory</i>
April 28, 2020	Working Group on Stratified Homotopy Theory, MSRI <i>The homotopy theory of stratified spaces</i>
April 24, 2020	Working Group on Stratified Homotopy Theory, MSRI <i>Intro to constructible sheaves & exit paths</i>
March 10, 2020	Pyknotic/Condensed Seminar, MSRI <i>Pyknotic spaces</i>

March 4, 2020	Pyknotic/Condensed Seminar, MSRI <i>Pyknotic sets</i>
December 2, 2019	Topology Seminar, Johns Hopkins <i>On the homotopy theory of stratified spaces</i>
October 13, 2019	AMS Fall Eastern Sectional Meeting, Binghamton, NY <i>On the homotopy theory of stratified spaces</i>
September 10, 2019	Topology Seminar, Notre Dame <i>Stratified spaces, constructible sheaves, & exit-paths</i>
August 8, 2019	Workshop on Homotopy Theory, Mathematisches Forschungsinstitut Oberwolfach <i>On the homotopy theory of stratified spaces</i>
October 18, 2018	Homotopy Harnessing Higher Structures Programme, Isaac Newton Institute <i>Constructible étale & analytic sheaves and exit paths</i>
December 7, 2017	Homotopy Theory Seminar, The Ohio State University <i>Monodromy & Stratified Homotopy Theory</i>

Teaching Experience

Spring 2026	Math 250B, Topics in Algebraic Geometry: Mirror Symmetry for Toric Varieties (graduate course), USC
Fall 2025	Math 226, Calculus III , USC
Spring 2025	Math 250B, Commutative Algebra (graduate course), UC Berkeley
Fall 2024	Math 256A, Algebraic Geometry (graduate course), UC Berkeley
Spring 2024	Math 250B, Commutative Algebra (graduate course), UC Berkeley
Fall 2023	Math 256A, Algebraic Geometry (graduate course), UC Berkeley
Fall 2019	18.821, Project Laboratory in Mathematics, Teaching Assistant , MIT

Mentoring Experience

Spring 2025	Undergraduate reading course on topos theory & categorical logic , UC Berkeley
Fall 2024	Undergraduate reading course on homotopy theory & higher category theory , UC Berkeley
Spring 2024	Undergraduate reading course on homotopy theory & higher category theory , UC Berkeley
Spring 2022	Undergraduate reading course on homotopy theory & higher category theory , UC Berkeley
2020–2021	Mentor , MIT Math Department's MIT Grad-Undergrad Math Mentoring Initiative (GUMMI) <i>Mentored two undergrads applying to math grad school</i>
2018–2019	Reading Course Mentor , MIT Department of Mathematics <i>Mentored an undergraduate reading course on differential forms and equivariant de Rham theory with Victor Guillemin</i>

Winter 2018	Directed Reading Program Mentor , MIT Department of Mathematics <i>Mentored an undergraduate in a category theory and topos theory reading course</i>
Summer 2017	$\sqrt{\text{Math}}$roots Academic Mentor <i>Mentored high school students through the MIT Math Department's $\sqrt{\text{Math}}$roots program</i>
Winter 2017	Directed Reading Program Mentor , MIT Department of Mathematics <i>Mentored two undergraduates in reading Emily Riehl's text Category theory in Context</i>

Service & Organizational Activities

Fall 2024	Faculty co-sponsor , MUSA 174, Student-taught course on category theory, UC Berkeley
2021–2022	Co-organizer , Nadler's Geometric Representation Theory Seminar
Fall 2020	Co-organizer , Harvard/MIT Thursday Seminar on Condensed Mathematics
Fall 2020	Co-organizer , MIT Juvitop Seminar on the Cobordism Hypothesis
2020–2021	Co-organizer , MIT Grad-Undergrad Math Mentoring Initiative (GUMMI)
Summer 2019	Co-organizer , MIT Reading Group on <i>An Inclusive Academy</i>
2019–2021	School of Science Graduate Student Council , MIT
Fall 2019	Co-organizer , MIT Juvitop Seminar on Differential Cohomology
Spring 2019	Co-organizer , Miniature Seminar on Factorization Homology
2018–2021	Coordinator , MIT PRIMES Circle
Fall 2018	Co-organizer , MIT Juvitop Seminar on <i>Ambidexterity in $K(n)$-local Stable Homotopy Theory</i>
Fall 2017	Organizer , MIT Topology Seminar
Spring 2016–2021	Diversity and Community Building Committee , MIT Department of Mathematics